

Background

A spring exerts a restoring force that is equal in magnitude but opposite in direction to the applied force:

$$F_x = -kx. \quad (1)$$

where k is the spring constant and has units Nm^{-1} . The term “restoring” force is used because it restores the spring back to its equilibrium (unstrained) position as soon as the applied force is removed.

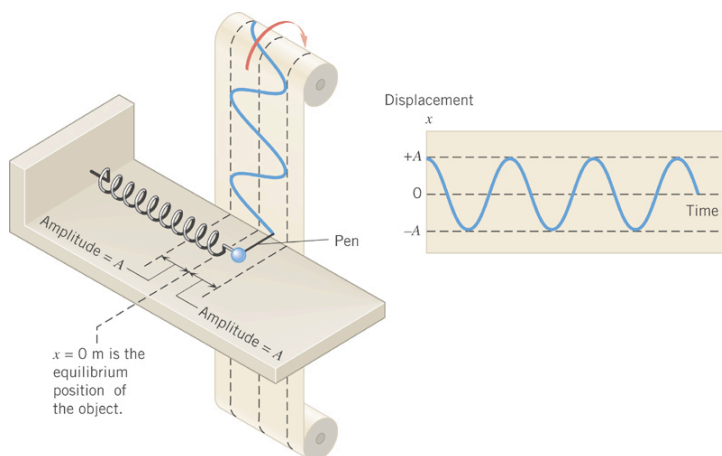
In an ideal environment (i.e. one without friction or air resistance) the spring would oscillate around the equilibrium position with oscillations of amplitude A . The displacement of a mass on the end of the spring would follow simple harmonic motion (see the figure below) and the displacement at any time t would be given by:

$$x(t) = A \cos(\omega_o t + \varphi), \quad (2)$$

where ω_o is the natural angular frequency of the mass on the end of the spring and φ some arbitrary phase offset. The period of the oscillations (in seconds) can be defined by:

$$T = \frac{2\pi}{\omega_o}. \quad (3)$$

where ω is related to the frequency f via $\omega = 2\pi f = 2\pi/T$.



In reality, the motion will be damped due to air resistance and the amplitude of the oscillations will decrease exponentially with time and the frequency of oscillation will also vary slightly from the natural frequency. In this case the motion can be described by:

$$x(t) = Ae^{-\beta t} \cos(\omega_1 t + \varphi), \quad (4)$$

where β is the damping factor, and ω_1 is the damped angular frequency.

Part 1: Determine the Spring Constant, k , using Hooke's Law

Using a single spring, hang a mass of known value m on the end of the spring and measure the spring's extension, x . Ensure you start with a mass that gives a measurable extension for the spring. Repeat the measurement for five different values of mass, each time determining the extension of the spring x from the unstretched position. **Note** – do not overstretch the spring ! Do not use any mass ≥ 500 g.

From Hooke's Law we have that the spring restoring force, F :

$$\begin{aligned} F_x &= -kx \\ \Rightarrow F_x &= mg = -kx. \end{aligned} \tag{5}$$

Plot a graph of the restoring force F_x against the extension of the spring x . The graph should yield a straight line with a slope equal to the spring constant, k for the spring you are using. From the graph determine k . Ensure your answer includes estimates for the maximum error on your value for k .

Part 2: Determine the Spring Constant, k , from the frequency of oscillation

The spring system consists of a magnetic (B-) field sensor, a magnet and a spring bar attached to the spring being tested. As the spring oscillates, so does the magnet, thereby changing the magnitude of the B- field measured by the sensor. The sensor is interfaced to the PC and a plot of B-field magnitude as a function of time (a time series) can be obtained. This software also enables you to plot the Fourier Transform (e.g. <http://mathworld.wolfram.com/FastFourierTransform.html>) of this time series: this allows you decompose the time series into a series of sinusoidal functions, giving the amplitude of each. In this way the strongest sinusoidal components present in the time series can be easily identified. (For this experiment this is done by clicking on the “open insert” tab, followed by “additional graph” and “FFT”). Hence we can use the Fourier transform to identify the frequency of oscillation of the spring. We will assume that this is the spring's natural frequency.

Procedure Hang a single mass onto the spring (50 g or 100 g is suitable) and measure the frequency of oscillation. Neglecting the mass of the spring and using $\omega_o^2 = k/m$, determine k for the spring. It is important that the spring only undergoes up-down motion. Comment on how the value you obtain using this method compares to that obtained in Part 1.

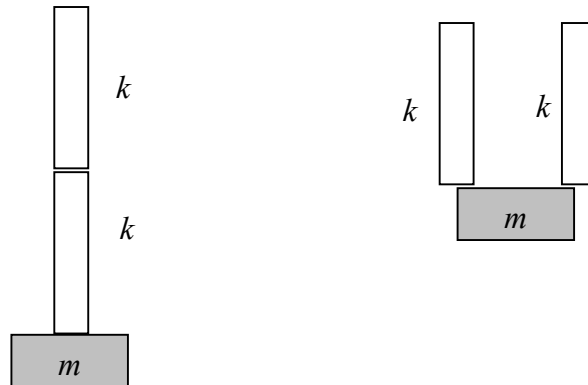
Note: make sure that only one pole rest under the sensor. For best results, use the negative pole.

Part 3: The damping of a system consisting of one mass and spring

Using a single mass, m , suspended on a single spring measure the damping factor β for the motion. For example, if amplitudes A_1 and A_2 are measured a time t apart, β can be calculated from $\ln(A_1/A_2) = \beta t$ (see Eqn 4). Comment on the value obtained.

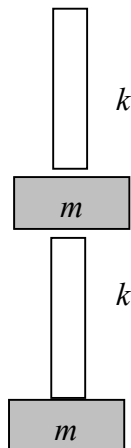
Part 4: Two springs and one mass: Springs in series and parallel

Determine the spring constants for the two possible spring combinations (by following the same method as given in Part 2) and compare with that for a single spring. Check the values you obtain against those expected from theory.



Part 5: Two springs and two masses: the motion of two coupled oscillators

When two equal masses and two identical springs are coupled as shown below, the motion appears to be quite complex.



In fact, this motion is a simple mixture of two fundamental modes of angular frequency ω_1 and ω_2 , where:

$$\omega_1^2 = \frac{3 + \sqrt{5}}{2} \omega_0^2 \quad \text{and} \quad \omega_2^2 = \frac{3 - \sqrt{5}}{2} \omega_0^2, \quad (6)$$

and $\omega_0^2 = k/m$.

Do the following:

- (i) Using two identical springs and two equal masses determine the two fundamental frequencies experimentally (again using the FFT). Compare these values to those expected from theory (equation 6).

- (ii) Use two springs and two different masses e.g. 50 g and 100 g. First determine the fundamental modes for the heavier mass on top and then the reverse. Do you get the same results?

Ensure you make a reasonable estimate of the maximum error in each of your measurements. All data that you record should show the measured value and its associated maximum error (and, of course, units).

References

1. "Physics" by Cutnell and Johnson.
2. "Classical Mechanics" by Kibble and Berkshire.